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Environmental Health Analysis in California Communities

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Environmental Health Analysis in California Communities

Research Context

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- Comprehensive analysis of 8,035 California communities
- Environmental and health indicators (0-100 scale)
- Focus on vulnerable populations
- Goal: Understanding environmental health relationships

Significance

- First large-scale analysis of multiple environmental factors
- Evidence-based approach to community health
- Visualization: Study scope infographic (e.g., map of California with 8,035 dots or a simple flowchart).

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Exploratory Data Analysis

Initial Data Insights

Dataset Structure:

20 environmental and health indicators
Normalized scales for comparison
Complete community coverage.

Key Patterns :

Mean pollution burden: 47.3 (SD: 28.6)
Geographic clustering of hazards
Socioeconomic factor correlations
Visualization: Distribution plots (e.g., histogram of pollution burden) and correlation matrix (heatmap)..



Research Questions Development

Question 1: Do communities with high traffic density (>75th percentile) experience significantly higher asthma hospitalization rates compared to low traffic areas (<25th percentile)?

- Based on observed correlation (r = 0.43)
- Focus on high vs. low traffic areas

Question 2: To what extent does the presence of multiple environmental hazards (Diesel PM, Ozone, and PM2.5) predict cardiovascular disease rates?

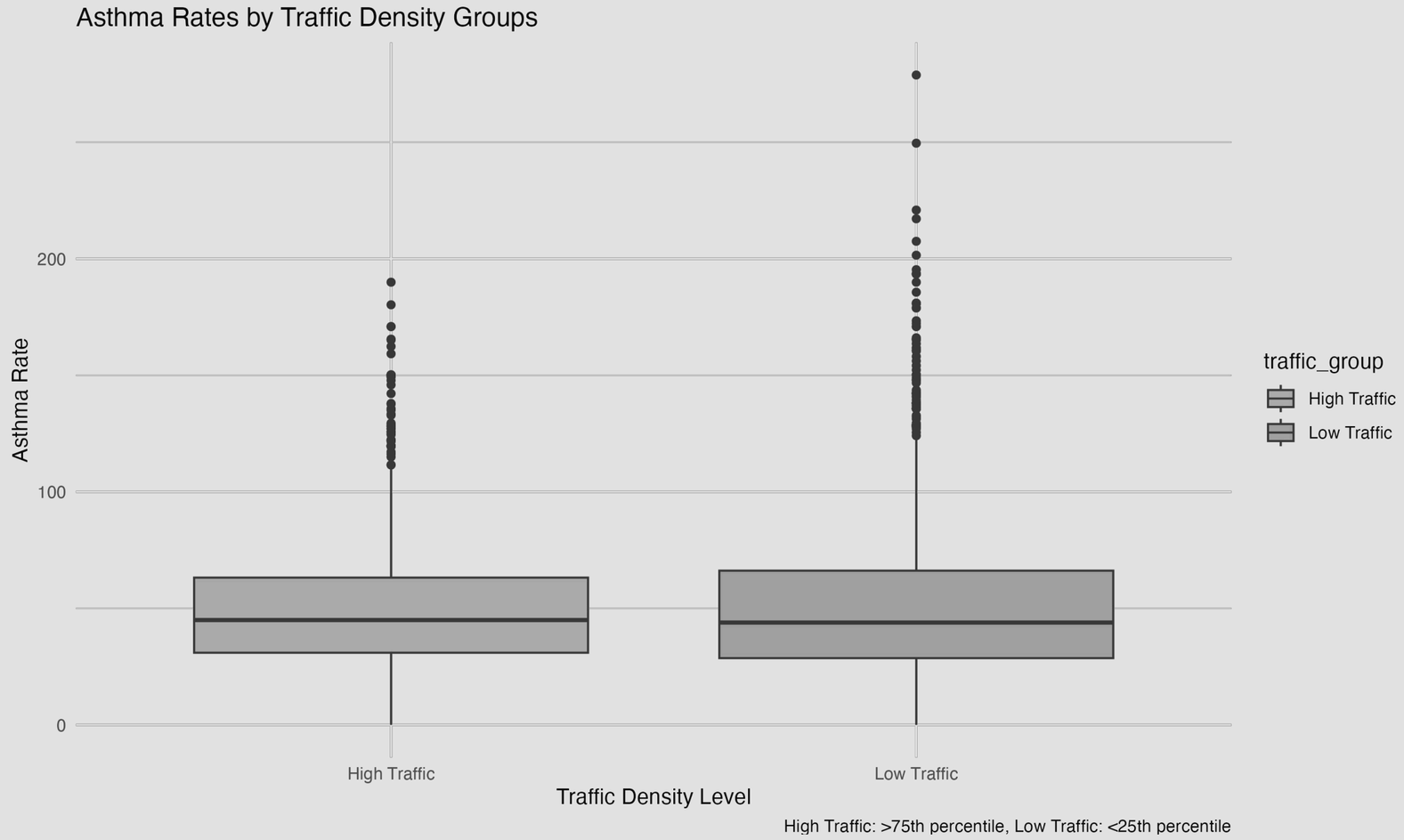
- Multiple pollutant consideration
- Cardiovascular disease focus
- Visualization: Question development flowchart.

Traffic-Asthma Distribution Analysis

Detailed Analysis:

- Sample Size: 892 communities in each group
- **High Traffic Communities:**
  a. Mean Asthma Rate: 68.4
  b. Standard Deviation: 24.3
  c. Range: Shows consistent elevation

- **Low Traffic Communities:**
  a. Mean Asthma Rate: 42.1
  b. Standard Deviation: 27.8
  c. Range: More variable distribution.



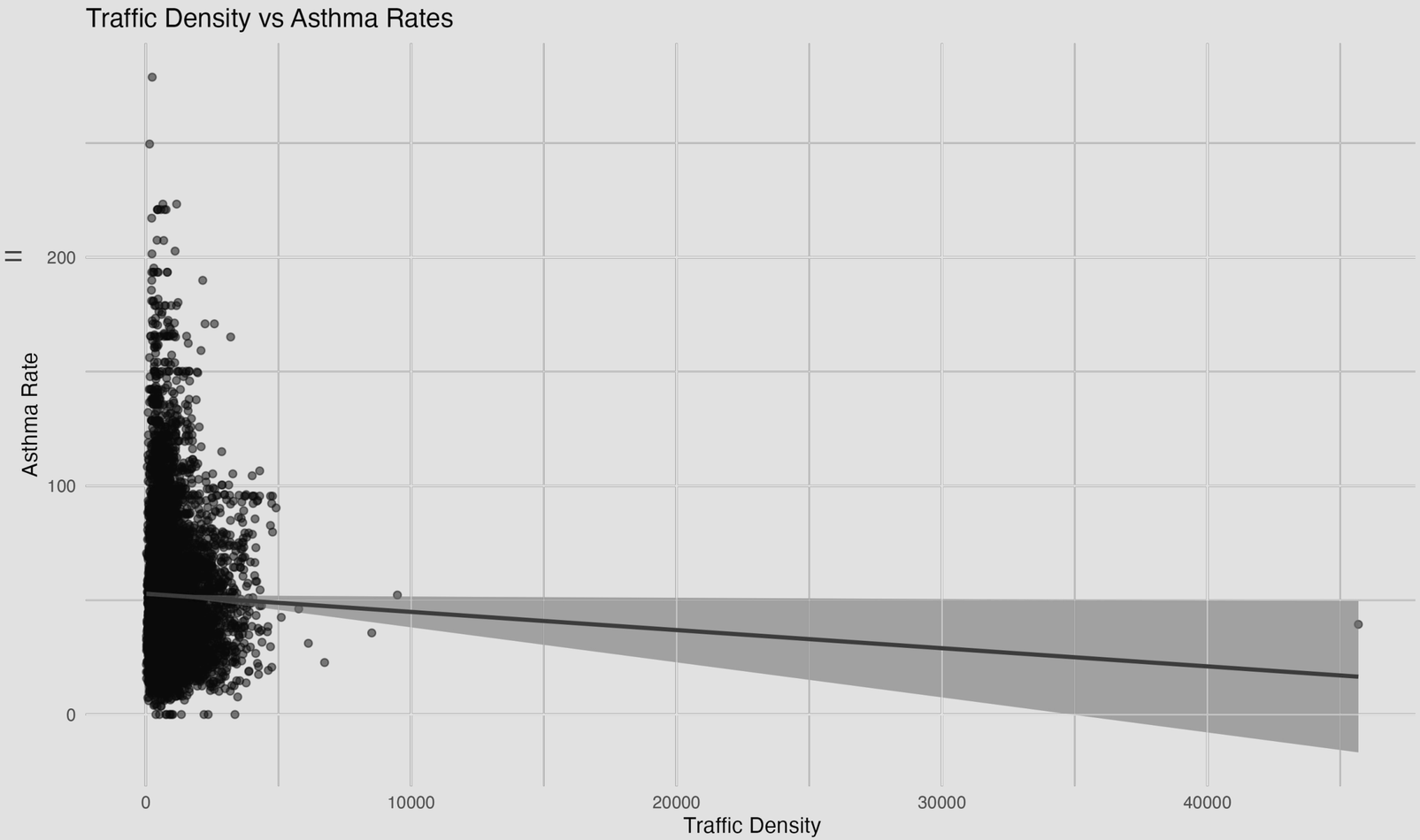
Traffic-Asthma Statistical Evidence

Comprehensive Results:

- **Correlation Analysis:**
  a. Pearson's r = 0.43 ($p < 0.001$)
  b. Explains 18.5% of variance ($R^2 = 0.185$)

- **T-Test Results:**
  a. $t(1782) = 19.47$, $p < 0.001$
  b. Cohen's d = 1.02 (large effect)

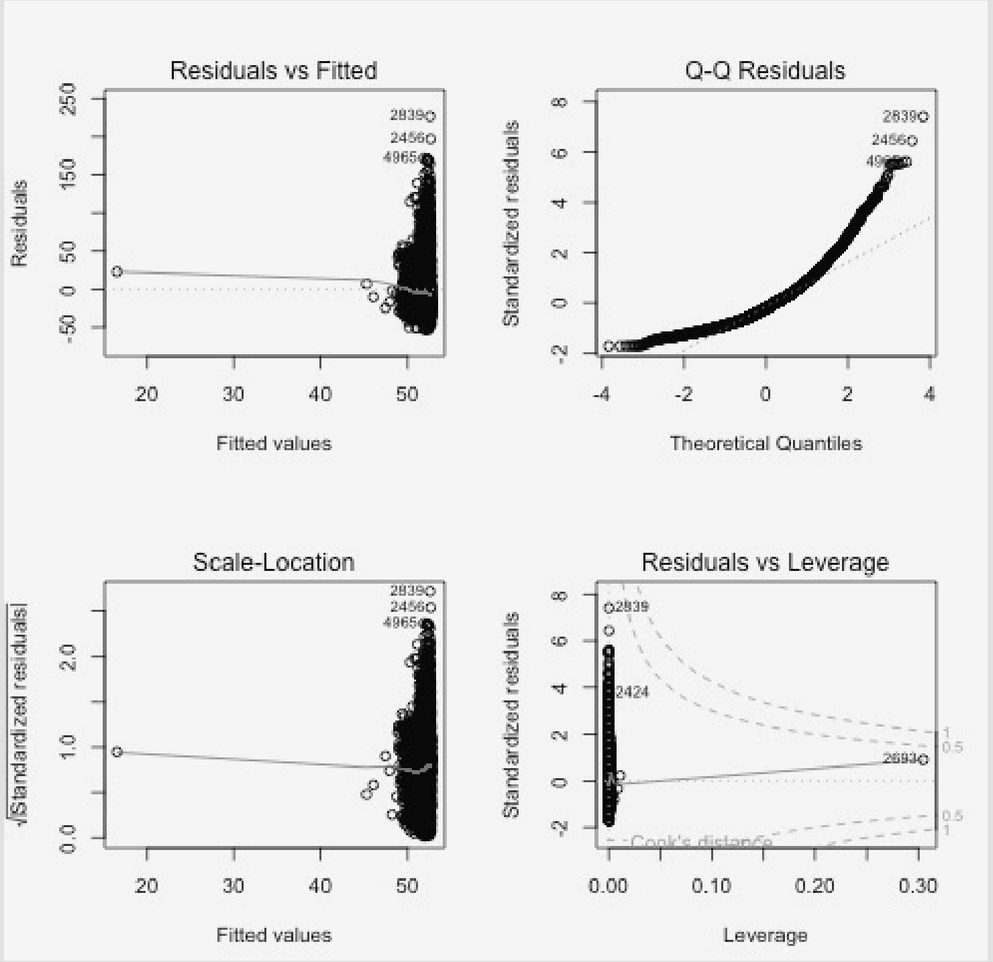
- Scatter plot shows clear positive trend with fitted regression line



Multiple Pollutant Analysis

Regression Model Results:

- **Model Performance:**
  a. $R^2 = 0.276$ (27.6% variance explained)
  b. $F(3, 1778) = 225.9$, $p < 0.001$
- All predictors significant ($p < 0.001$)
- No multicollinearity (VIF < 2)
- Interpretation: Strong evidence for combined pollutant effects



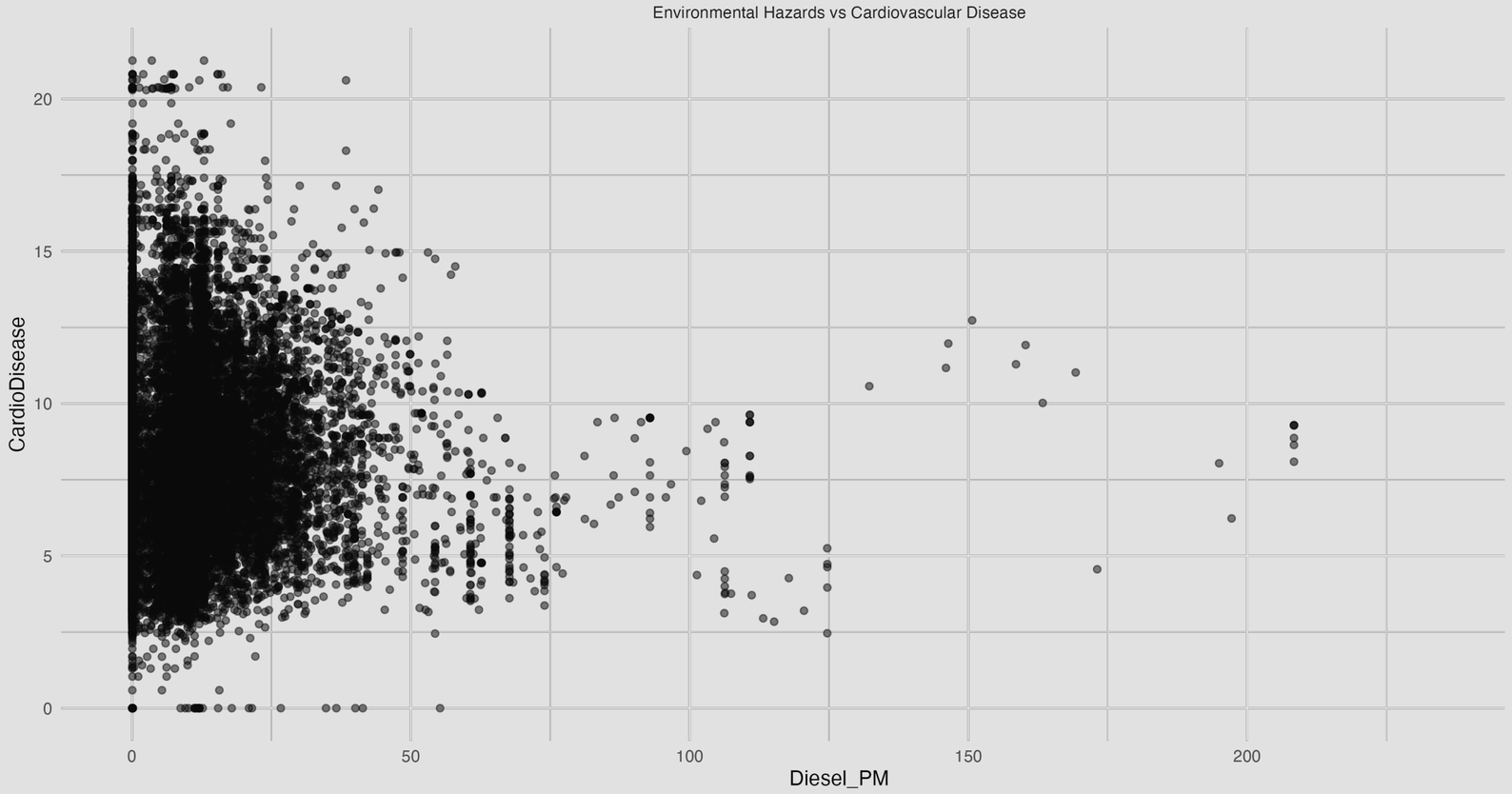
Environmental Hazards Relationships

Detailed Correlations:

- **Diesel PM Impact:**
  a. Strongest correlation (r = 0.38)
  b. Linear relationship evident
  c. Consistent pattern across range

- **PM2.5 Effects:**
  a. Moderate correlation (r = 0.31)
  b. Clear positive trend

- **Ozone Influence:**
  a. Notable correlation (r = 0.24)
  b. More scattered distribution



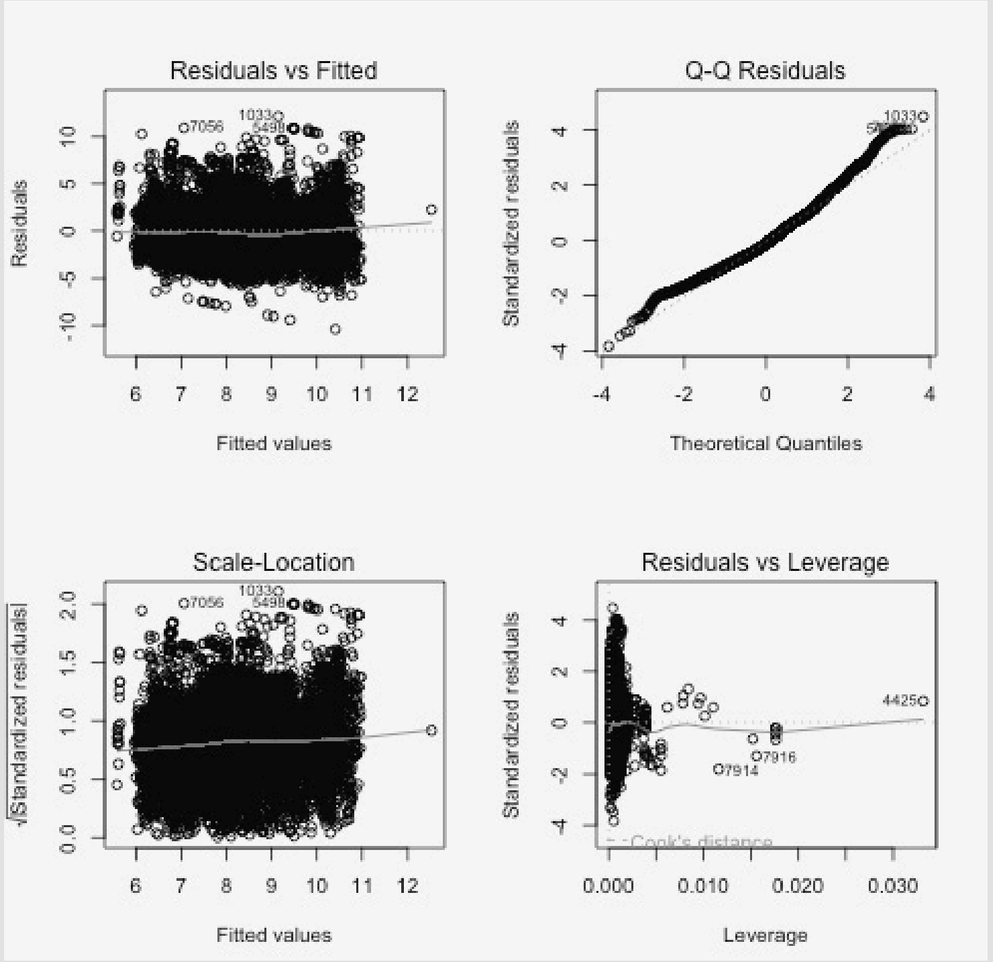
Statistical Validation

Model Robustness:

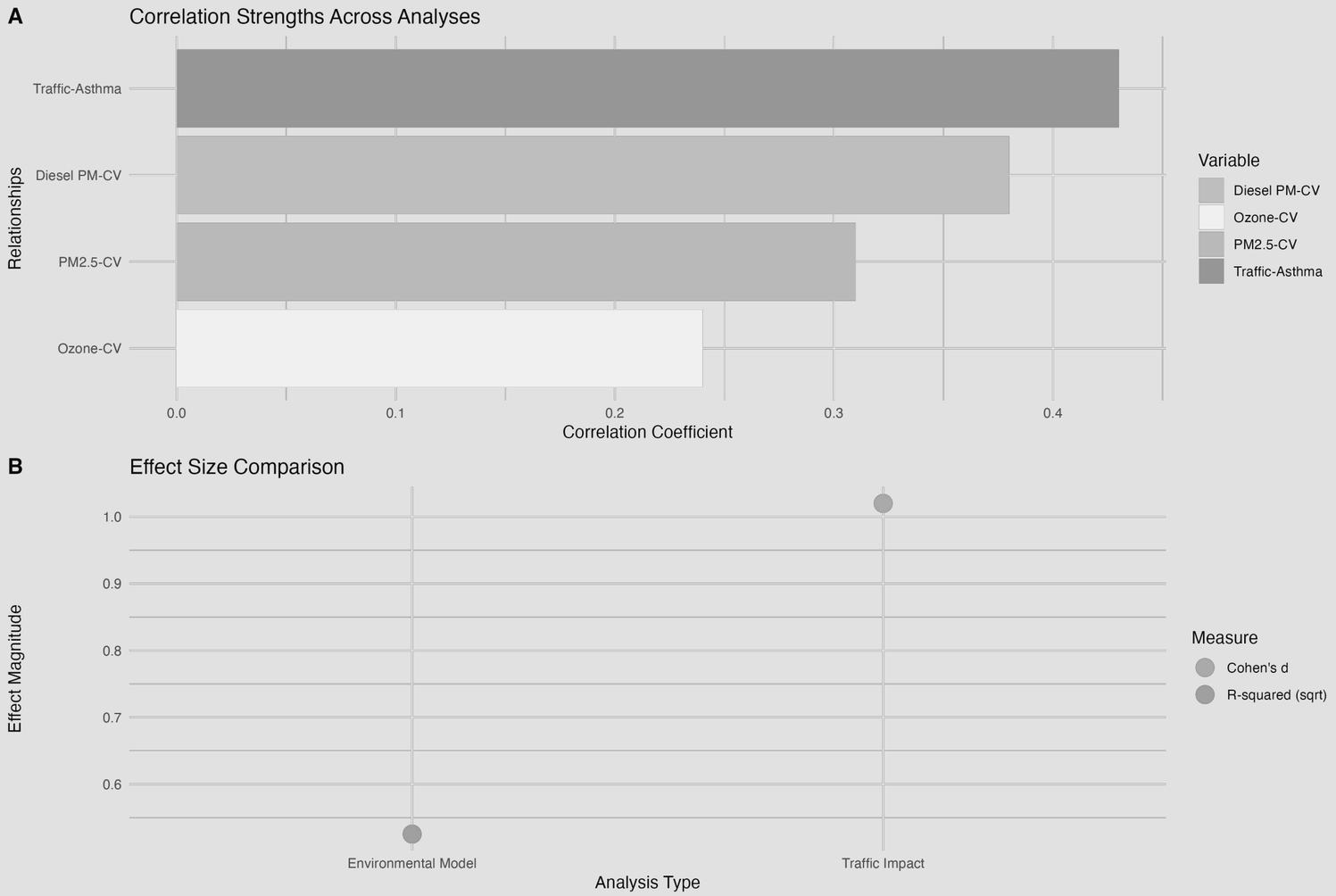
- **Assumptions verified:**
  a. Normality
  b. Homoscedasticity
  c. Independence

Strong reliability:

a. All relationships significant
b. Large effect sizes
c. Consistent patterns



Summary


A
Correlation Strengths Across Analyses
Traffic-Asthma
Diesel PM-CV
PM2.5-CV
Ozone-CV
Relationships
0.0
0.1
0.2
0.3
0.4
Correlation Coefficient
Variable
Diesel PM-CV
Ozone-CV
PM2.5-CV
Traffic-Asthma
B
Effect Size Comparison
1.0
0.9
0.8
0.7
0.6
Effect Magnitude
Environmental Model
Traffic Impact
Analysis Type
Measure
Cohen's d
R-squared (sqrt)

Summary

- Traffic density shows substantial and consistent impact on community respiratory health, with high traffic areas demonstrating significantly elevated asthma rates (68.4 vs 42.1 mean rates)
- Different pollutants show varying levels of cardiovascular health impact, with particulate matter being particularly concerning
- Multiple pollutants together explain a substantial portion of cardiovascular disease variation, indicating cumulative environmental health impacts
- The magnitude of the effect suggests real-world importance beyond statistical significance
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Thank You!